

How money moves

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UCT Financial Innovation Hub Bootcamp

From the history of money to the art of making payments

- Yesterday: money has three functions; financial intermediaries facilitate payments — and **gate access** to them
- The running question of this course: **who gets to participate in the financial system?**
- Today's version of it: **who can access the payment rails — and who is locked out?**
- Today: how are payments made? What does the payment infrastructure look like? How are cross-border payments made?
- *Payment rails are not neutral plumbing — who may hold a settlement account, how settlement is designed, and how cross-border chains are built all decide who can pay, and at what cost*

Balance Sheets

Balance sheets, in one slide

- A payment is, in its essence, a **change to balance sheets** — value moving across accounts.
- A balance sheet is a snapshot of what an entity *owns* (**assets**), *owes* (**liabilities**), and the residual *equity*
- It must always balance — the accounting equation:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

- Double-entry bookkeeping (codified by Pacioli, 1494) is what keeps it balanced — a foundational financial *technology* that made firms legible to lenders¹. We won't drill the debit/credit rules — what matters for payments is the **balance sheet** it produces

¹Goetzmann, *Money Changes Everything*: accounting as a core financial technology.

A bank's balance sheet

Assets	Equity and Liabilities
• Reserves • Cash • Securities/Bonds • Loans • Company • Consumer • Real estate • Other • Other assets	• Deposits • Debt • Inter-bank loans • Central bank loans • Other • Shareholder equity

Your deposit is the bank's IOU

- Your money in the bank is the bank's *liability* — its **IOU** to you
- You don't "have money at the bank" — the bank **owes** you money
- A payment from you to me is the bank reassigning whom it owes — moving an IOU across its books
- "Settlement" is just moving these IOUs across *banks'* books²
- This single idea is what the whole settlement section builds on

²Graeber, *Debt*: money is largely an entry on a balance sheet — a credit/IOU relationship, not a thing.

A balance sheet is information — and information is the price of entry

- A balance sheet is **information** — exactly what a lender needs to judge a borrower
- Recall Lecture 1: asymmetric information → adverse selection → lenders **ration credit** → missing markets
- Most firms in developing economies — and most South African MSMEs — are **informal**: they keep no formal books, so they are **invisible to lenders** and excluded from credit, even when profitable
- **The record is the barrier.**

Digital payments create the record

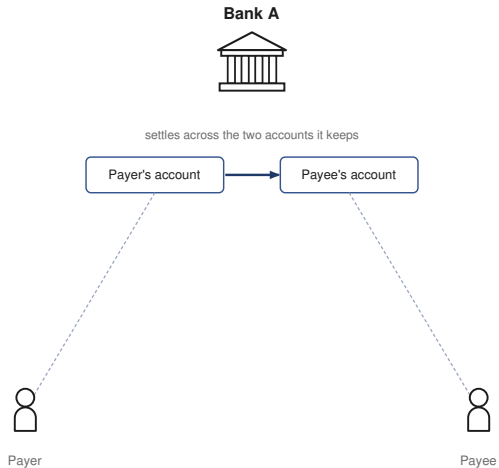
- Mobile-money and card transaction histories **substitute for formal books** — they make the excluded *visible*
- M-Shwari and Tala score credit straight from M-PESA and airtime data; card/POS adoption pulls small firms toward formality³
- A recurring idea in this course: **the payment rail is also a data rail** — paying digitally is how many of the excluded enter formal finance

³Suri (2017); Bharadwaj & Suri (2020); Higgins (2024). We return to this in Lecture 3.

A worked example: a deposit

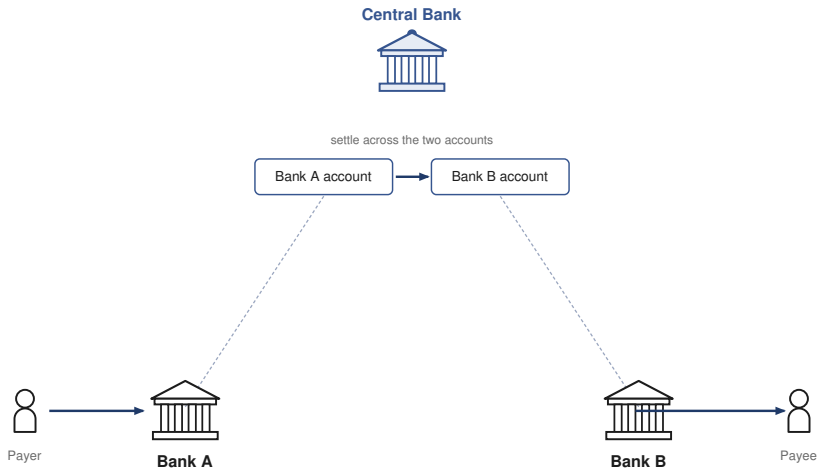
- Thandi deposits R1 000 in cash at Bank A
- Bank A's balance sheet moves on **both** sides:
 - **cash / reserves (asset)** + R1 000
 - **Thandi's deposit (liability)** + R1 000
- $A = L + E$ still holds – the two sides move together
- The deposit Bank A now records is its **IOU** to Thandi. Next: what happens when Thandi *pays* someone?

A payment within one bank



Nothing leaves Bank A — it just reassigns whom it owes

A payment between two banks



Settlement systems

Money enables payments

- Yesterday we introduced the concepts of money and financial intermediaries
- We said that a key role of money is to facilitate transactions by making it easier to buy and sell goods
- Put differently, money enables payments
- Today we turn our attention to how payments work and how financial intermediaries facilitate these payments
- We'll cover a number of examples, each increasing in complexity.
 - Starting with a basic payment between two parties all the way through to cross-border payments

Payments

- Payments and payment systems are the plumbing of the economy
- For most of modern history, payments were boring and uninspiring
- This has changed in the last 50 or so years: debit cards, mobile payments, crypto, CBDCs
- Payments are big business
 - In 2022, \$8.7 trillion worth of digital payments made
 - In 2022, payment revenue accounted for 36% of total bank revenue

Payment system

Changes in the payments ecosystem have ushered in four eras of payments business models.

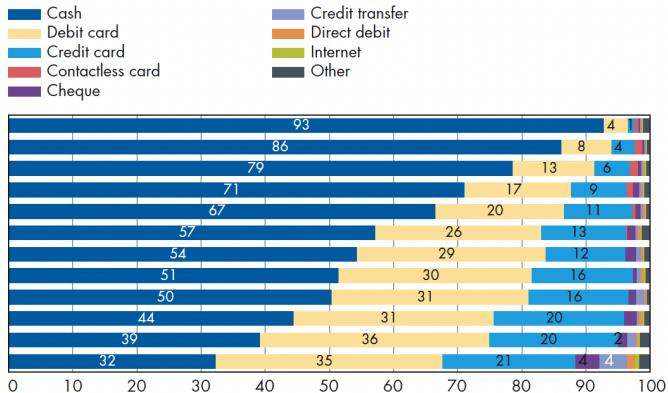
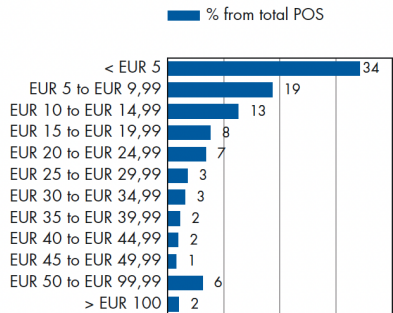
Timeline



A horizontal timeline arrow pointing from left to right, with tick marks for the years 1960, 1970, 1980, 1990, 2000, 2010, and 2020. The arrow is colored with a gradient: dark blue for the 1960s, light blue for the 1970s-1980s, medium blue for the 1990s-2010s, and green for the 2020s.

	 Paper Era: Pre-1960s	 Plastic Era: 1960s–90s	 Account Era: 1990s–2020s	 Decoupled Era: 2020s
Transactions	Cash, checks, and wire transfers	Cash, checks, wire transfers, and physical cards	Instant transfers, A2A, and virtual cards	Interoperable and open, platform, and decentralized
Sources of economic differentiation	Balances and deposits	Transaction fees	Relationships and transfer fees	Convenience, security, and low fraud incidence
Distribution channels	Physical (eg, branches)	Physical and ATMs	Physical, ATMs, online, mobile, and digital wallets	Physical, ATMs, online, mobile, embedded, and metaverse
Technology	Telegram	Automated Clearing House (ACH)	Applications and instant payments	Platform as a service (PaaS), tokenization, generative AI, and open/API banking

How do people pay



The interbank funds transfer systems

- In the classic model of a financial system, banks facilitate payments
- We call the system banks use to transfer money, the **interbank funds transfer system**
- Two types of systems
- (1) Wholesale funds transfer systems
 - Infrequent, low volume, large and time critical payments
 - Mainly financial market transactions
- (2) Retail transfer systems
 - Frequent, large volume, small payments
 - POS transactions, EFTs, cheques etc.

Key concepts in interbank funds transfer systems

Transfer of information and processing

- A funds transfer is proceeded by a message, called a **payment order** requesting the transfer of funds to the payee
- These are typically credit transfers - both payment order and funds is sent from the bank of the payer to the bank of the payee
- However, they can also take the form of debit transfers (e.g debit orders)
- Once payment order is recieved, it is **processed** - identity, account and funds verification. Also referred to as **clearing**
- Once cleared, these payment orders represent a binding payment obligation

Key concepts in interbank funds transfer systems

Settlement

- Once a payment order has been received and cleared, settlement can take place
- Once settlement is initiated, it is irrevocable and unconditional
- Settlement typically takes place in one of two places
- A transfer across the books of a bank via a **clearing house** in the case of retail transfers (cleared with **commercial-bank money**)
- A transfer across the books of the **central bank** in the case of wholesale transfers (cleared with **central-bank money**)
- *Who may hold central-bank money (reserves)?* Only banks. Everyone else holds **commercial-bank money** and must transact *through* a bank

Settlement systems

- Interbank funds transfer systems can be classified in several ways
- Settlement systems differ in *when* and *how* they settle
 - When? At the end of each day (**designated-time/deferred**), or after each transaction (**continuous**)?
 - How? Settle net balance (**net**) or settle each transaction (**gross**)?

Settlement characteristics	Gross	Net
Designated-time (deferred)	Designated-time gross settlement	Designated-time net settlement (DNS)
Continuous (real-time)	Real-time gross settlement (RTGS)	Not applicable

- For many years, most banks utilized a **DNS** settlement system
- Today, **RTGS** settlement systems are commonplace

Net vs gross: same logic, different timing

Gross — RTGS (real-time)

reserves move on every transaction



2 transfers → 2 reserve moves

Net — DNS (deferred)

reserves move once, at day-end



2 transfers → 1 net move

same balance-sheet logic — the only difference is WHEN reserves move

The advantages of net settlement

- Easier liquidity management
 - Netting of transactions requires smaller net payments and less capital/liquidity
- Reduces the number of transfers
- Delegating clearing and settlement allows for multilateral settlement
 - Transactions can be netted across multiple institutions resulting in fewer transfers
- Transaction can be reversed within a window of time
 - As settlement happens at a given time each day, transactions can be reversed before then

The disadvantages of net settlement

- Transactions take time to clear
- Complex clearing, especially for multilateral settlement
 - Scope for errors
- Silo record keeping
- Concentrates settlement risks
 - Settlement risk concentrated at a single point in the day
 - If issues arise, little time to address them

The advantages of gross settlement

- Faster transactions
- Reduces settlement risk
 - Real-time settlement restricts settlement risk to the point of transactions
- Safety - transactions are no longer grouped together
 - Large grouped payments pose a greater cybersecurity risk

The disadvantages of gross settlement

- More transactions to process
- Amplifies operational or technical failures
 - Downtime or interruptions to the payment system can be destabilizing
- Introduces liquidity risk
 - Institutions need to have sufficient capital on hand to settle at any point

Who is allowed into the core?

- Direct settlement access — an account at the central bank — has historically been **bank-only**
- So fintechs and mobile-money providers must **ride on banks' rails**: they rent access from a sponsor bank, which raises cost and lets the bank gatekeep
- The design of who-may-settle is therefore a **financial-inclusion** choice, not a technical detail
- South Africa is only now opening National Payment System access to **non-banks** (“proportional” regulation) — widening the door to the core

Cross-border settlement

Global transfers and settlement

“The holy grail of cross-border payments is a solution allowing cross-border payments to be

- immediate,
- cheap,
- universal, and
- settled in a secure settlement medium.

The search for such a solution is as old as international commerce and the implied need to pay... after more than thousand years of search, the holy grail of cross-border payments can be found within the next ten years” - European Central Bank (2022)

Challenges around global transfers

“Cross-border payments sit at the heart of international trade and economic activity. However, for too long cross-border payments have faced four particular challenges:

- high costs,
- low speed,
- limited access, and
- insufficient transparency.

Faster, cheaper, more transparent and **inclusive** cross-border payments would have widespread benefits for supporting economic growth, international trade, global development and **financial inclusion.**” - Financial Stability Board (2021)

Remittances: the human stakes of cross-border

- The typical use case: migrant workers sending money home
- > US\$650 bn flowed to low- and middle-income countries in 2023 – **larger than foreign direct investment and aid combined** for many of them.⁴
- More stable and counter-cyclical than other flows – in Lesotho, The Gambia and Comoros, remittances are **over 20% of GDP**
- So this is the **retail, human end** of the payment system – and it is exactly where the system works worst

⁴World Bank/KNOMAD; IOM (global formal remittances rose from \$128 bn in 2000 to \$831 bn in 2022).

The cost and who pays it

- Sub-Saharan Africa is the **most expensive region on earth** to receive money: sending \$200 costs $\approx 8.8\%$, against a $\approx 6.5\%$ global average⁵
- And it depends on *how* you send it:
 - **Banks** $\approx 14.5\%$ · MTOs $\approx 5\%$ · **mobile money** $\approx 3.6\%$; digital $\approx 4.9\%$ vs non-digital $\approx 7.2\%$
- The targets: **SDG 10.c** — under **3%** by 2030; the **G20** — no corridor above 5%
- A 7% fee on a R1 200 transfer is a day's food taken from the household that can least spare it

⁵World Bank *Remittance Prices Worldwide*, Q1 2025.

Why so expensive? The system is built for the big players

- The global cross-border market approached $\approx$ **US\$1 quadrillion** in 2024 — but it is concentrated in **advanced economies, financial institutions and large transactions**⁶
- Frictions and information costs fall hardest on **small, retail** payments — remittances are a structural *afterthought*
- Every hop in the correspondent chain (recall Figure 3) adds a fee and an FX margin — the **chain is the cost**
- Fixing it is not charity: it is redesigning a wholesale system so the smallest payment is served as well as the largest

⁶Cerutti, Firat & Hengge (IMF, 2025), using Swift data.

South Africa to SADC, up close

- South Africa is the region's remittance **hub**: formal outflows to SADC **tripled** — R6 bn (2016) to **over R19 bn (2024)**⁷
- Zimbabwe is the dominant corridor; the top four (Zimbabwe, Lesotho, Malawi, Mozambique) are $\approx 90\%$ of formal flows
- ≈ 4 million SADC migrants in SA, $\approx 89\%$ **undocumented** — so they fail KYC at the **first mile** and are pushed to cash and informal channels (80–90% of high-volume corridors)
- Think of the chain as **first mile** (sender access) → **middle mile** (cross-border infra) → **last mile** (payout). For inclusion, the binding constraint is usually the **first mile** — exactly where de-risking bites next

⁷FinMark Trust / SARB, *South Africa to the rest of SADC Remittances Market Assessment* (2024).

High costs and low speed. Why?

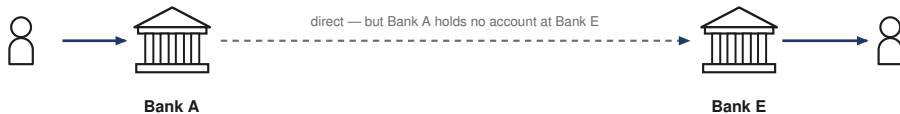
- Long transaction chains
- Currency conversion
- Time zone differences
- Exchange controls
- Anti-fraud and anti-money laundering

Long transaction chains

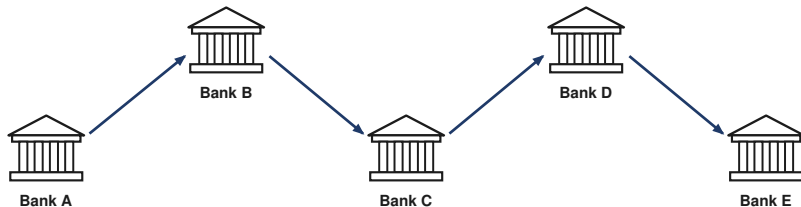
- Many settlement systems require a number of parties involved in order to enact a payment
- Cross border settlements have typically been enacted via nostro and vostro accounts
- Nostro account → a bank account that a domestic bank has with a foreign bank in foreign currency (*our* money, held there)
- Vostro account → the same account seen from the other side (*your* money, held by us)
- Each hop in the chain is just **one more account-keeper** — and when no single keeper connects both ends, the chain lengthens

Long transaction chains: the route you want vs the route you get

What you want

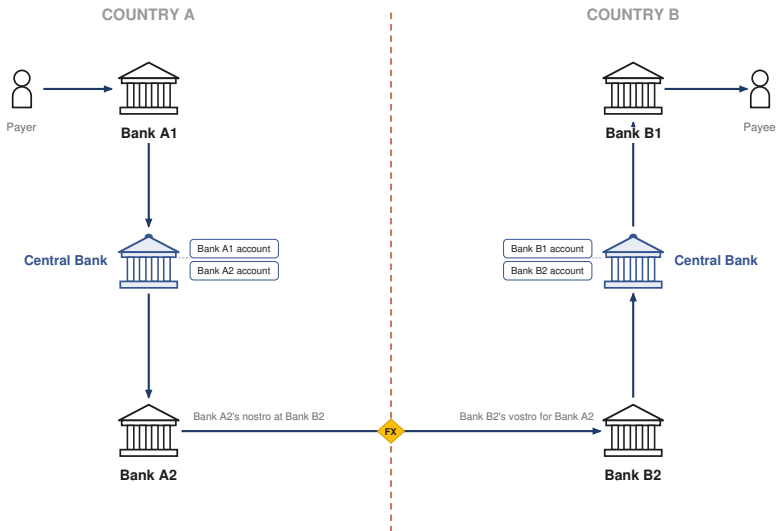


What you get



every hop is a correspondent bank — the chain is the cost

The long transaction chain (correspondent banking)



Time zone differences and currency risk

- Every example so far assumed Country A and Country B were in the same time zone
- But imagine two countries where the overlap in bank operating hours is small (1–2 hours)
- This increases processing time and *currency risk*: value has left one side but not yet arrived at the other
- Question of **authority** (who handles the conversion) and **timing** (when it occurs)

Exchange controls

- → government-imposed restrictions on the purchase and sale of currencies
- Imposed in many countries in the world, especially in emerging markets
- In such countries, all incoming and outgoing payments require the payer and payee to report the reason for the payment to their respective central bank
 - In SA, for example, Regulation 3(1) of the Exchange Control Regulations prohibits any person from transferring funds out of South Africa without the approval of the SARB
- This can lead to significant delays, especially for larger amounts

Anti-fraud and anti-money laundering

- Two features of cross-border payments that induce significant regulatory costs
- (1) AML/CFT → Anti-money laundering and countering the financing of terrorism
 - Financial Action Task Force (FATF) → international regulatory watchdog that oversees compliance with anti-money laundering rules
 - In the news: South Africa was **greylisted** in February 2023 and **exited the greylist in October 2025**⁸
- (2) KYC → Know your customer requirements
 - Requirement for financial institutions to verify the identity of their customers and the legitimacy/legality of their payments and source of funds

⁸FATF; South African National Treasury media statement, 24 October 2025.

Searching for the holy grail of cross-border settlement

Towards the holy grail

- “after more than thousand years of search, the holy grail of cross-border payments can be found **within the next ten years**” - European Central Bank (2022)
- We'll look at 6 types of settlement systems that hold promise. The unifying question: **who is the shared account-keeper?**
 - Correspondant banking
 - Using a subsidiary FX bank
 - Regional RTGS
 - FinTech payment providers
 - Crypto, Stablecoins
 - CBDC

Correspondant banking

- The 'conventional' model: transfer of funds via nostro accounts, requiring an internationally active correspondent bank (the chain on the previous figure)
- Innovation of instant payments has made this process significantly faster and removed settlement risk

Pros and cons of correspondent banking

- Pros

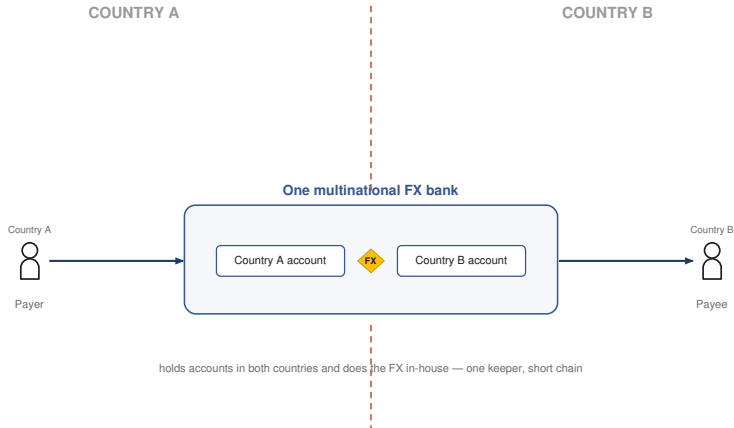
- Well-established (banking networks, kyc)
- Diversifies the number of payment providers (increased competition)
- Has universal reach

- Cons

- Many intermediaries → slow and expensive
- Transactions clear in commercial money and not, central bank money
- When, and who to handle forex conversion?

↪ *Inclusion*: de-risking excludes whole regions of poor countries — the chain is where inclusion breaks.

Using a subsidiary FX bank



Pros and cons of subsidiary FX banks

- Pros

- Well-established (banking networks, kyc)
- Simpler than correspondent banking
- Subsidiary bank handles forex conversion

- Cons

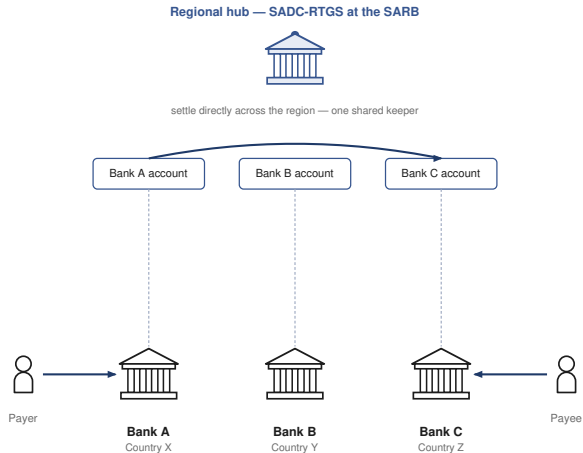
- Requires multinational FX banks
- Concentrates settlement risk in a small number of FX banks

↪ *Inclusion*: simpler, but access is concentrated in a handful of multinationals — thin corridors stay unserved.

Regional RTGS

- In certain scenarios, customers may prefer to settle payments in currencies other than their own. Why?
 - Firms who conduct majority of their business in another country
 - Weak local currency
 - Dominant currency in a region (e.g South African Rand)
- This has led to the creation of Regional RTGS, managed by one central bank
 - E.g. SADC-RTGS
 - SARB manages clearing and settlement and allows banks in SADC to hold Rand accounts at the SARB

Regional RTGS: one shared hub, no chain



Pros and cons of regional RTGS

- Pros

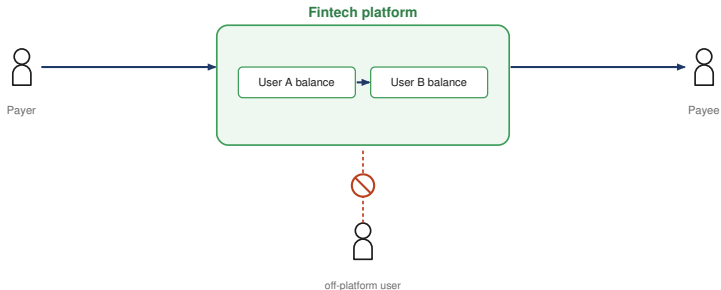
- Settles in central-bank money — no correspondent chain, low cost, fast
- Brings a whole region onto one rail

- Cons

- Built around a dominant currency and a single hub central bank
- Non-members are still outside

↪ *Inclusion*: includes the region — but around the Rand and the SARB; it deepens, rather than removes, hub dependence.

FinTech payment providers



closed loop — users hold value with the fintech; anyone off-platform is excluded

Pros and cons of FinTech payment providers

- Pros

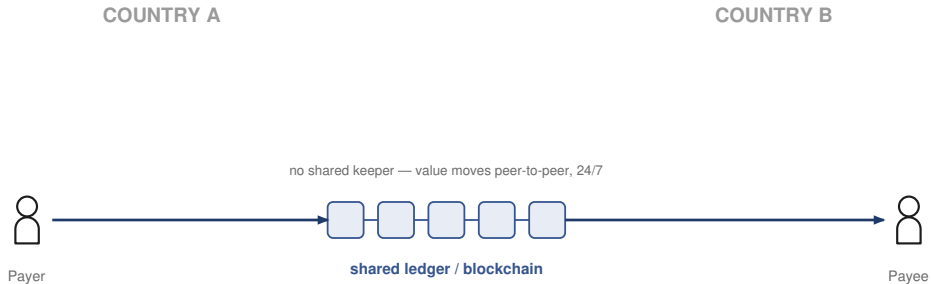
- Many providers → competition drives down costs
- Especially efficient if the FinTech does not hold deposits
- Often specialized services for specific financial services
- Low overheads compared to banks

- Cons

- Can low-fee income model be sustainable?
- Often non-interoperable, closed loop solutions

↪ *Inclusion*: cheap and competitive, but closed loops exclude anyone off the platform — interoperability is the inclusion test.

Crypto and stablecoins



variants: (1) crypto (2) cashed in/out via a bank account (3) stablecoin = reserve-backed token

Pros of crypto and stablecoins

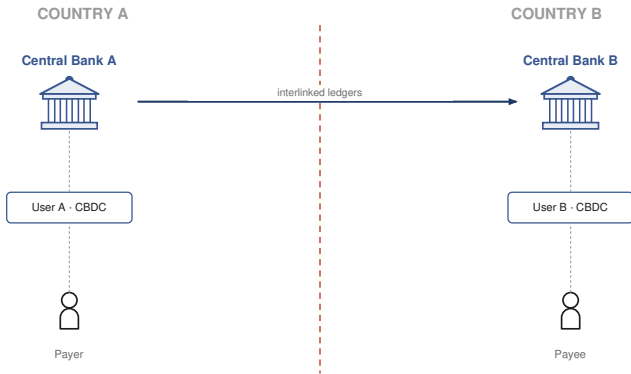
- Decentralized
 - No intermediaries
 - Efficient — payments clear in a fraction of the time
 - Low to no fees — lack of intermediation reduces fees
 - Currency agnostic — removes currency risk
 - 24/7 → removes time zone risk
- ↪ *Inclusion*: bypassing banks is potentially inclusive — reaching anyone with a phone and a wallet.

Cons of crypto and stablecoins

- AML/CFT concerns
 - Lack of regulatory oversight
 - Does crypto meet the definition of money? → price volatility
 - Threats to monetary sovereignty
 - Financial stability issues
 - Large, safe reserves required to back a stablecoin; managing them is hard
- ↪ *Inclusion*: but volatility, device/connectivity barriers and AML gatekeeping can re-exclude the very people it promises to reach.⁹

⁹Rogoff, *The Curse of Cash*: cash remains the inclusive default, and the state will keep control of the unit of account.

Central bank digital currencies (CBDC)



non-banks hold central-bank money directly — no commercial-bank account needed

Pros and cons of central bank digital currencies (CBDC)

- Pros

- Removes the need for a bank account
- Settles in central-bank money; non-banks can hold and convert it
- Cross-border via direct central-bank interlinks — no correspondent chain

- Cons

- Requires CBDC uptake on both sides
- Design questions: privacy, disintermediation of banks, offline access

↪ *Inclusion*: directly inclusive in principle — “no bank account needed” — but only if uptake and access (devices, connectivity, ID) actually reach the excluded.

Scoring the six on inclusion

Solution	Cost	Speed	Who's in	Who's still out
Correspondent	high	slow	incumbents	de-risked poor countries
Subsidiary FX	med	fast	multinationals' clients	thin, unserved corridors
Regional RTGS	low	fast	the region	non-members; hub-currency dependence
Fintech	low	fast	on-platform users	anyone off the closed loop
Crypto/stablecoin	low	fast	the connected	no device/ID; AML-blocked
CBDC	low	fast	anyone (no bank a/c)	those without up-take/access

Have we found the holy grail?

Have we found the holy grail?

- The holy grail will likely feature three solutions
 - Subsidiary FX banking (high fees!)
 - Stablecoins (high reserves required! regulatory concerns!)
 - CBDCs (uptake? high fees!)
- Key question: can we remove fees?
 - Fees are driven by costs of intermediation across border, e.g forex
- And a fifth criterion the ECB list omits: **inclusive**. The grail must reach the unbanked migrant, not only the corporate treasurer

Key takeaways

- Payment rails are not neutral: who may hold a settlement account, how settlement is designed, and how cross-border chains are built decide **who can pay** and **at what cost**
- A deposit is the bank's IOU; a payment is those IOUs moving across books — **on-us**, via the **central bank**, or down a **chain** of keepers across borders
- The chain is the cost: cross-border friction — de-risking especially — falls hardest on the poorest, making reform an **inclusion** issue
- Digital payments also **create the records** that pull the informally excluded into formal finance
- *Recurring framework: What problem does it solve? Who is excluded, and why? Access vs. use? Who gains and who loses? Role of the state?*